Psychiatric Quarterly (2023) 94:33–47
https://doi.org/10.1007/s11126-022-10007-6

ORIGINAL PAPER

Adaptation and translation of a scalable measure for assessing health worker competence to deliver a brief psychological intervention: A case study from central India of its comparison to a performance-based measure

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Accepted: 12 November 2022 / Published online: 20 December 2022
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Abstract
Efforts to reduce the global burden of common mental disorders have focused on scaling up evidence-based training programs for non-specialist providers to deliver brief psychological interventions. To evaluate these provider training programs, appropriate and scalable assessments of competency need to be developed alongside them. We followed a systematic approach for the cultural adaptation and translation into Hindi of a valid, English, multiple-choice applied knowledge measure to assess non-specialists' competence to deliver a brief psychological intervention for depression in rural India. We then explored the relationship between the performance of 30 non-specialist providers on the same written measure compared with a structured performance-based measure consisting of two role-plays. The results of the multiple-choice assessment had an overall mean score of 37.40 (SD = 11.31) compared to the mean scores of role-play A (the easier role-play) of 43.25 (SD = 14.50) and role-play B (the more difficult role-play) of 43.25 (SD = 13.00). Role-play performance-based measures and written applied knowledge measures represent different approaches with unique strengths and challenges to measuring competence. Scaling up training programs requires the development of scalable methods for competency assessment. Exploring the relationship between these two measures, our team found no apparent differences between the two modes of assessment. Continued comparison of these approaches is needed to determine the consistency of outcomes across the two formats and to link the scores on these measures with clinical performance as reflected by the quality of care and patient outcomes.

Trial Registration: ClinicalTrials.gov Identifier: NCT04157816; 8th November 2019.

Keywords Depression · Health worker competence · Cultural adaptation · Translation · Mental health · Training · Psychological intervention · Global mental health · Task sharing

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Introduction

A substantial and rapidly growing body of evidence over the last three decades has shown that when adequately trained and supervised, non-specialist providers (NSP's) can effectively deliver psychological interventions for a range of common mental disorders including depression, anxiety, and other trauma-related disorders in low, middle and high-income countries (HICs) [1–5]. NSP's may include community health workers, primary care workers, midwives, traditional healers, peers, teachers, and other lay persons who do not have specialized training in mental health care. These psychological interventions for common mental disorders are recommended by the World Health Organization (WHO) [6] as the first line of treatment; however, such interventions are inaccessible to most individuals who need treatment [7].

A major barrier to implementing sustainable task-sharing initiatives is a lack of culturally appropriate and easily administered tools to support training and assessments of clinical skills and competencies for their effective delivery [8]. Assessments of competence are an important tool for supporting implementation, as they can be used to inform the quality of different training programs, tailor supervision, identify health workers in need of additional support, and address gaps in knowledge and skills through supplementary training [9]. Psychometrically validated and easily administered assessments for measuring competency need to be developed and implemented alongside training programs to effectively build workforce capacity to deliver evidence-based brief psychological interventions for mental disorders and to scale up and sustain these interventions within routine care settings.

Achieving competence, a required outcome of training has been defined as attaining knowledge of this specific form of psychological intervention as well as an ability to apply this knowledge, skillfully and effectively, in practice [10]. There is currently a gap in the evidence base as to what methods of assessment are most scalable and appropriate to use to assess NSP's' ability to deliver brief psychological interventions. Even in professional psychology programs, no single strategy to assess the competency of providers has been proven to be more resource-efficient or to adequately cover the broad range of competencies [11].

Role play-based assessments using 'actors' following a standardized script to simulate a patient interaction is a method widely used as a training outcome for providers [11]. This method includes identifying the competencies to be assessed, developing a scenario and script, having individuals act out the scenario, and then having an expert rate an individual's ability to apply the skills they learned in the training program in this simulated clinical environment [11]. Role play based assessments following a training program have a number of strengths including the promotion of active learning, the ability to address 'red-flags' in provider's delivery of care, and simulation of a relationship between provider and patient [12]. Role play scripts can also be tailored to multiple applications such as training evaluation, clinical supervision, selection for training or intervention, and as a measure of fidelity in intervention trials [13]. While there are many strengths to this method of assessment, there are also a number of challenges including the need for financial and logistical resources to implement trained actors, conducting and rating the role play scenarios at scale, issues of inter-rater reliability, limited assessment of a few competencies, and the artificiality of the situation [10–12, 14].

Written assessments are another method used as a training outcome for providers. These may include multiple choice, matching, fill-in-the-blank, and true-false questions based on vignettes or problem-solving scenarios [11]. Written with the aim of assessing applied knowledge of the treatment intervention, these scenarios require integrative problem-solv-

ing and decision making that can evaluate the essential elements of a broad range of core and treatment specific competency domains [11]. Another strength of written assessments is that they are particularly beneficial when trying to scale up training programs to reach larger populations because they are inexpensive, easy to implement and are a method that allows for repeat testing of training outcomes over time (e.g., before training, after training and at periods thereafter) [14] and assessing multiple people at the same time [11]. However, written assessments have their own challenges, which include concerns about not actually assessing performance and the predictive validity of performance in professional practice settings [11]. Despite these challenges, written assessments represent a potentially highly scalable approach for assessing competency, as these tools can be administered remotely using digital devices or via paper and pencil for individuals who may have limited experience using digital technologies. It is likely that the optimal approach for comprehensive assessment of competency will require a combination of these different strategies, at different time points following training, employing both written assessments and role plays.

As part of a larger program of research [15] aimed at scaling up task sharing of a brief psychological intervention for depression in primary care settings in rural India, we developed and validated a brief, scalable multiple choice measure of competence [14] for assessing NSP's ability to deliver a brief behavioral activation treatment for depression called the Healthy Activity Program (HAP) [16]. The multiple choice applied knowledge measure was validated, using Rasch analysis, as being of suitable level of difficulty for NSP's and other lay providers [14]. The process of development and validation of this measure followed prior research showing that these assessments are a reliable tool to assess therapist competence following training [10, 17]. The HAP manual and the multiple choice applied knowledge competency measure were initially developed in English [14], thereby limiting application across settings where English is not the primary language.

Just as it is necessary for psychological treatment interventions and the training curriculum to deliver these interventions in a culturally adapted way to meet the needs of the local context, customs, and language [18, 19], training outcome measures also need to be appropriately and adequately adapted and translated while retaining the core content and competencies that they were intending to assess in their original development. Best practice for translation includes parallel translation, team discussions, extensive quantitative and qualitative pretesting with respondents, and detailed documentation [24, 25, 35]. In this study we describe the process our team used to translate and adapt the HAP multiple choice assessment measure of applied knowledge for assessment of NSP competency in rural India.

As efforts to train NSP's to deliver psychological interventions across a wide range of settings continues to gain attention, it is imperative that consistent, culturally appropriate, valid, and scalable measures to assess the outcomes of these training programs are developed and evaluated. Thus, the purpose of our study was two-fold. First, to describe the systematic approach to the cultural adaptation and translation into Hindi of a valid, English, multiple-choice applied knowledge measure to assess NSP competence to deliver a brief psychological intervention for depression. Second, to explore NSP's' performance on the adapted multiple-choice applied knowledge measure compared with a standardized structured role-play performance-based measure.

Methods

Study Setting and Participants

This study was embedded in a three-arm randomized pilot study [20]. The NSP's were recruited for this study from April to August 2019 in three community health centers in the Sehore district of Madhya Pradesh. Sehore is one of the five districts (out of 51 districts) in Madhya Pradesh, one of the largest states in central India where the National Mental Health Program (NMHP) is operational. The NMHP was launched in India in 1982 in accordance with the WHO's recommendations to ensure the availability and accessibility of minimum mental health services for all under the framework of general health care system in the community [21]. Sangath, the research organization leading this project, has a close partnership and an established Memorandum of Understanding with the state government and thus the reason why this study site was selected. All procedures described in this study were approved by ethical review boards at Sangath, India (VP_2017_028) and Harvard Medical School, USA (IRB17-0092).

The target sample of NSP's consisted of 3 cadres of frontline health workers, including Accredited Social Health Activists (ASHAs), ASHA facilitators, and multi-purpose workers (MPWs). These frontline health workers are trained and employed with the National Health Mission, working primarily to deliver essential routine primary care services and to promote utilization of the existing health services.

Training Programs & Intervention

All NSP's participating in this study completed training in the delivery of HAP, a brief evidence-based and manualized behavioral activation intervention consisting of 6 to 8 sessions for patients with moderately severe to severe depression (assessed by the Patient Health Questionnaire-9 (PHQ-9) score >= 10). Recommended by the WHO as part of the Mental Health Gap Action Programme (mhGAP) [6], HAP consists of core and treatment specific strategies including psychoeducation, behavioral assessment, activity monitoring, activity structuring and scheduling, and activation of social networks. The manual is designed for delivering treatment in primary care settings by lay counsellors who do not have formal training in mental health [16]. The training for NSP's was designed specifically for this cadre of health workers to gain clinical skills and competencies necessary to deliver HAP for treatment of depression [20]. It was delivered either through conventional face-to-face classroom-based instruction, or as part of a digital training program accessible by smartphone application [20]. The content of the digital training program was the same as the conventional face-to-face training [22].

Part 1: Adaptation of a Multiple-choice Competency Assessment Measure

The first aim of the study involved the cultural adaptation and translation of a brief scalable applied knowledge measure previously developed by our team to measure provider competence in HAP. Detailed methods for the development and initial validation of the English version of the measure are described elsewhere [14]. Three psychometrically sound versions of the measure of equivalent difficulty measuring common general counselling skills, understanding of logistics

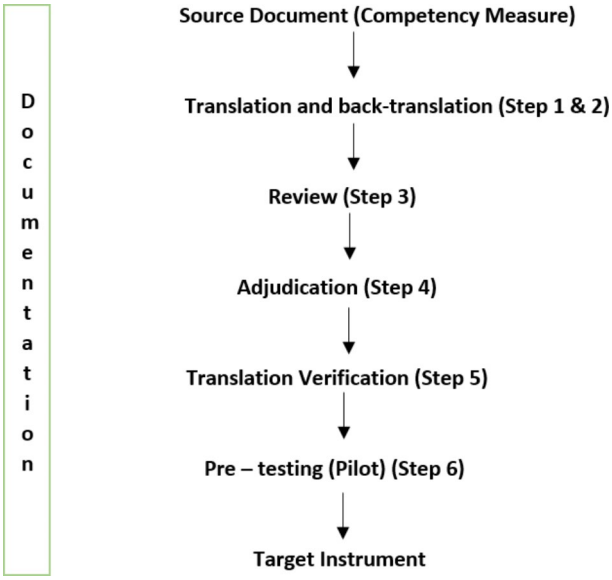


Fig. 1 Translation, Review, Adjudication, Pretesting, and Documentation (TRAPD) methodology

for session delivery, and HAP treatment specific skills were derived to allow for repeat testing of training outcomes over time. Each test includes 26 one-best-answer multiple choice items. The items use a short real-world scenario clinical vignette as the item stem, have a lead-in question and four answer options. Each item is weighted equally with higher scores on the overall measure reflecting higher levels of knowledge and competency.

To use the measure to assess the competence of the NSP's participating in the HAP training in Madhya Pradesh, it was necessary to produce an accurate translation and adaptation of the original English version of the measure into the Hindi language. To achieve this goal, we followed an iterative process informed by the TRAPD model (Translation, Review, Adjudication, Pretesting, and Documentation) [23], which focuses on cultural equivalence rather than on literal equivalence at the level of words or entities [23]. Importantly, the TRAPD model advocates for establishing an adaptation group with diverse expertise; translation and checking for accuracy of language; identifying local terms for the conditions of interest; adaptation of materials to fit local idioms, images and circumstances, while maintaining essential components; and consultation to reach consensus [23]. This approach is a directive of the World Health Organization - World Mental Health (WHO-WMH) consortium for cross sectional surveys and has been widely used by the WHO and European social surveys [25, 26]. Figure 1 outlines the steps used to produce the final Hindi translation using the TRAPD model.

Three different English versions of the measure were translated into Hindi (forward translation) during Step 1 by independent translators fluent in both Hindi and English. The translators were tested for their translation skills (i.e., linguistic and cultural knowledge) on a sample (10%) of the source document. A member of the review team (UJ) assessed the translator's level of skill to ensure the quality of translation, whether the translator was suitable for the project and to identify any serious deficiencies in the translation. Following this, two bilingual psychologists from the Sangath Delhi office carried out an initial review of the translation to overcome potential limitations (regional variance, idiosyncratic inter-

Table 1 Socio-Demographic Characteristics of NSP's

Participant Characteristics	n (%)
n	30
Gender	
Male	1 (3.3)
Female	29(96.7)
NSP Designation	
ASHA	19 (63.3)
ASHA Facilitator	10 (33.3)
Multi-purpose Workers	1 (3.3)
Education	
8th standard	11 (36.7)
9th -10th standard	7 (23.3)
12th standard	6 (20.0)
Degree or higher	6 (20.0)
Experience as an NSP	
Years, mean (95% CI) *	6.9 (5.5, 8.5)
Age	
Years, mean (95% CI) *	34.1 (31.6, 36.6)

pretations) of working with only one translator. The reviewers carefully reviewed usage of technical terms, inserted notes, and corrected or suggested replacement words, considered to be more suitable to convey the intended meaning of the original text.

A similar approach was employed for back-translation in Step 2. Back translation means that the translation which has been produced from the target language is re-translated or translated back into the source language. This process was carried out to find any discrepancies in the target language text [25]. A bilingual, independent, non-mental health professional was specifically hired for this work to back translate the measures into English. Following the back translation, for Step 3 of the process, two of the research team's native English speakers then independently reviewed the back translated versions with the original source document to assess its appropriateness and equivalence to the original measure. These two researchers were involved in the development of the English measure, but not involved in the translation process prior to this stage.

Step 4 involves adjudication, which includes a final decision on translation options in cooperation with a reviewer. The purpose of the adjudication process was to assess the accuracy of the Hindi translation as well as its intended meaning in English. Consistency checks were performed by a team member (UJ) to look for use of terms, rendering of words, spelling, and punctuation. This ensured harmony in translation, and discrepancies identified were resolved in consensus with the reviewer. Subsequent to this process, modifications were incorporated based on consensus and the measure was finalized for translation verification.

Step 5 included verifying the translations before pre-testing to minimize idiosyncrasies, identify errors, assess language understanding and comprehensibility with a group of NSP's (consisting of six ASHAs and one ASHA facilitator) who were involved in the formative stages of the intervention development process. Based on group discussions, the translation was revised to eliminate potentially remaining discrepancies between understanding of the item and its intended meaning. In Step 6 the revised measure was pilot tested with participants enrolled in a randomized pilot study [20] to test the viability, and address any remaining errors of the transla-

tion in preparation for a large randomized controlled trial [15]. Following this pilot testing, we arrived at the target instrument.

Part 2: Exploring Performance on the Multiple-choice Measure Compared to Standardized Role-play Assessment

Our second aim was to explore the use of the multiple-choice measure in comparison to a standardized structured role-play performance-based measure. We used existing standardized role-plays which were previously developed for the prior trial assessing the effectiveness of HAP (called the PREMIUM trial) that was conducted in Goa, India [27, 28]. The role plays covered Phases 1 and 2 of HAP and were translated from English to Hindi for use in the current study. The standardized role-plays included a description of clinical situations to be addressed, a summary of patient's progress in the treatment and some additional information to better understand patient circumstances. It also consisted of a written account of each scenario for actors describing how a patient should generally respond and how a patient should respond in a particular circumstance depending on the scenario.

To examine the NSP's performance on the standardized role-play measure, we administered the role-plays to NSP's after completion of their HAP training [20]. Each NSP was provided two randomly assigned hypothetical counselling sessions, one relatively easy scenario (i.e., Role Play A) and the other somewhat difficult scenario (i.e., Role Play B). This same approach using an easier and more difficult scenario was previously employed in the PREMIUM trial [27]. Each scenario involved the NSP playing the role of a "therapist" treating a "patient", played by an actor. The NSP's were given five minutes of preparation time and were asked to capture a short sample of treatment. The scenarios lasted no longer than 10 min. The role-play scenarios were videotaped for the purpose of external rating and assessment.

Rating Procedure for Standardized Role-Play Assessments

Each NSP completed a pair of role-plays (i.e., one easy and one difficult). A NSP was randomly allocated among 4 raters, such that each pair of role-plays was rated by at least 2 raters. The independent raters were 4 experienced HAP counselors who had not participated in prior phases of this study and were blind to the NSP's' identity and training status. These experienced HAP counselors evaluated the videotaped role-plays. To minimize potential selection bias, the NSP's were randomly assigned to the 4 raters by a member of the research team who was independent of the data collection, rating process, and blind to the names/identity of the raters or NSP's. The raters were also provided with a copy of both actor and counsellor scripts of each role-play for reference. The steps for rating the role-play videos are outlined below.

The role-plays scenarios are accompanied by a procedural checklist which provides scenario specific guidance specifying both core and desirable features, an important guide for rating. The procedure checklist consists of three different sections (Core Features, Conditional Core Features, and Optional Features) designed to provide guidance to the rater as to the relevant procedure-specific HAP competencies for the session that were met and were not met.

After using the checklist as a guide, the raters then were required to make a judgement on the quality of the implementation of the NSP's performance during the session using a 4-point Likert scale. The scale ranged from a score of 0 (*complete absence of HAP procedures and/or red flags*) to a score of 3 (*consistent application of HAP procedures i.e., all*

scenario specific core features present and no red flags). The score of 1 indicated *limited or inconsistent application of HAP* (i.e., few of the scenario specific features were present and/or potential red flags) and a score of 2 indicated *moderate application of HAP procedures* (i.e., most of the scenario specific features were present and no red flags). "Red Flags" were defined as potential concerns raised while performing a role play such as undermining/inconsistent statements and/or NSP's performing well on the overall procedure but also performing badly in some aspects of counselling (e.g., unwelcoming/non empathic style, non-adherent to treatment style, etc.). All 4 raters reviewed and rated the role plays independently for participant 1 and participant 2 and then discussed their scores to obtain a consensus global score. The goal of this initial step was to train all raters in using the rating scale and to ensure that the scoring was completed in the same way by each of them.

For participants 3–10, each rater independently scored 4 role plays (2 raters for each role play). Raters then met with their partnered rater to discuss their ratings on the 4 role plays. They retained and discussed their independent scores and came to a consensus for an additional global score for each role play. The raters did not change their independent scores after the discussion. This step was repeated for participants 11–20 and participants 21–30 with the raters discussing their scores in between these batches of participants to reach consensus for a global score for each role play.

Data Analysis

For the rating of the standardized role-plays, we calculated a one-way random effects model, average measures intraclass correlation coefficient (ICC). ICCs were calculated in a grouped manner as follows: participants 1–2 only, then participants 3–30. This approach was followed to establish reliability among the 4 raters first on the initial 2 test participants before proceeding with the full number of participants 3–30.

Values of n (%) and mean (SD) were used as appropriate to describe the sociodemographic characteristics, competency assessment scores and role-play scores of the participants collected after completing the HAP training program. For each individual NSP, the applied knowledge competence scores were compared with the ratings on the role-play measure to determine whether competence on the former (i.e., a multiple-choice applied knowledge measure) was consistent with competence on the latter (i.e., standardized role-play performance-based measure). We calculated Pearson correlations for the scores of the multiple-choice assessment compared with the scores on the role-play assessments (separate correlations calculated for role play A and role play B). The scores for the multiple-choice assessments and the role-play assessments were standardized on a scale of 0-100 to ease comparison. Summary statistics and correlations were generated using STATA version 14.0 (StataCorp).

Results

The final Hindi adapted version of the multiple-choice measure included three psychometrically sound tests of equivalent difficulty, and each with 26 items. The procedures for administration of the Hindi translated multiple-choice applied knowledge competency assessment to the NSP's in the pilot study is described in detail elsewhere [20].

Participant Characteristics

A total of 30 NSP's who participated in the training programs completed the post-training multiple-choice competence assessment and the standardized role-play assessments. The mean age of the 30 NSP's was 34.1 (SD = 6.75) years, and nearly all participants (29 out of 30 NSP's) were female. Their mean number of years in their current role was 6.98 (SD = 4.01) (Table 1).

Inter-rater Reliability of the Role-play Assessments

For participants 1–2, the estimated correlation between individual ratings was 0.58, indicating moderate similarity between ratings within a target. The estimated intraclass correlation between ratings averaged over k = 4 judges was higher, 0.85. The average ICC will typically be higher than the individual ICC. The estimated intraclass correlation measures the similarity or reliability of mean ratings from groups of four judges. We do have statistical evidence that either ICC is significantly different from zero based on reported confidence intervals and the one-sided F test. For participants 3–30, the estimated correlation between individual ratings was 0.56 (95% CI = 0.35-0.72), indicating moderate similarity between ratings within a target. The estimated intraclass correlation between ratings averaged over k = 2 judges was again a moderate value, 0.72 (95% CI = 0.52-0.83). This analysis is based on a one-sided F test with F (55.0, 56.0) = 3.56 and Prob > F = 0.0003.

Comparing the Multiple-choice Assessment Scores and the Standardized Role-play Assessment Ratings

The results of the multiple-choice assessment had an overall mean score of 37.40% (SD = 11.31; Min = 19.1, Max = 77.2), whereas the mean scores of role-play A and B were 43.25% (SD = 14.50; Min = 25; Max = 75). These scores are aggregate post-training scores for participants, where the scores are presented as percentages with higher scores reflecting higher levels of knowledge and competency. The range of scores appear to be similar for the validated multiple-choice assessments and the standardized role-play assessment scores. The interquartile range for the competency assessment score was 14.73 and for both sets of role-play scores it was 25. Figure 2 reflects consistency across the mean scores for the three assessments of competency. The role play A score and the multiple-choice assessment were found to be moderately positively correlated, $\rho = 0.36$ (p-value = 0.0495). The role play B score and the multiple-choice assessment were also found to be moderately positively correlated, $\rho = 0.30$ (p-value = 0.1019).

Discussion

Scaling up training and supervision methods for NSP's to deliver brief psychological interventions is required to address the unmet needs for mental health care in areas that otherwise have had limited access. Innovative, scalable approaches to workforce training, such as use of digital technology for the HAP training program [22], need to be complemented with culturally appropriate, scalable, and valid measures to assess the provider competence following training in order

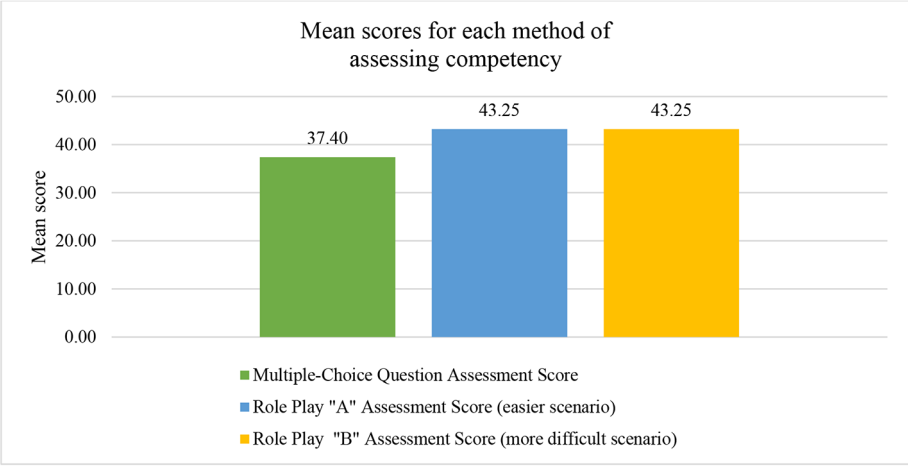


Fig. 2 Mean scores for the applied knowledge multiple choice assessment and the two role play scenarios

to achieve quality treatment and sustainable systems of care in communities [14, 29]. This study describes two important processes for achieving these goals. First, our team successfully utilized a systematic approach to support the cultural adaptation and translation of an English, scalable, multiple-choice applied knowledge measure into Hindi for assessing NSP competence to deliver HAP in Madhya Pradesh, India. Second, we found that the relationship between the NSP's' performance on the Hindi version of the multiple-choice measure appeared consistent with an existing standardized structured role-play performance-based measure, which is the more labor-intensive form of assessment. Therefore, the culturally adapted multiple choice questionnaire not only appears comparable to the existing gold standard of structured role-play measures of competency, it is also considerably easier to scale and administer in the routine settings where health workers are usually trained.

Translation and Adaptation

There are several existing frameworks to cultural adaptation which includes surface compared to deep structure adaptations (i.e., modifying superficial materials such as illustrations or language to better fit preferences of beneficiary population, instead of conveying salience of an intervention by considering how the particular target population perceive the content) [30]; core compared to peripheral data driven cultural adaptation framework to distinguish between therapeutic ingredients that cause symptom change rather than peripheral aspects which are related to acceptability and feasibility of an intervention [31]; cultural concepts of distress, which builds on dimensions of surface and deep structure adaptation, yet there is a lack of theoretical underpinnings on research standards to evaluate the quality of translation in cultural adaptation literature [32].

We followed the multi-step translation methodology of the TRAPD model [23] to complete the cultural adaptation and translation of the multiple-choice applied knowledge assessment [14]. Our method aligns with similar approaches to deep structure adaptation and cultural concepts of distress, as it fosters empirical research including a human cen-

tered approach, and transparent criteria on how to implement and document the process of cultural adaptation. As researchers and program implementers plan to translate and adapt measures for various contexts, this model is important to follow as it goes beyond the need for translation from one language to another but emphasizes the importance of making sure the essential components of the measure are preserved while adapting the measure so that it is culturally appropriate for the context it is expected to be used in.

Comparing the Multiple-choice Assessment Scores and the Role-play Assessment Ratings

The written applied knowledge measure represents a dramatically different approach to measuring health worker competence compared to existing role-play performance-based measures; yet, both offer unique strengths as well as challenges. Importantly, despite the advantages or disadvantages with each form of measurement, it is recommended, in the early stages of developing these measures, to use a combination of assessment tools to enhance validity of the assessment measures [11]. By exploring the relationship between the NSP's' performance on the multiple-choice applied knowledge measure with the standardized structured role-play performance-based measure, our team aimed to determine whether there might be similarities or differences in performance among NSP's across both measures.

In comparing the two modes of competency assessment our team found no apparent differences between the two modes of assessment in our study. While the NSP's' scores appeared consistent, similar on both the multiple-choice measure and the standardized role-play measure following completion of the training, none of the NSP's scored above the fiftieth-percentile on either. Participants in this study were consistent in their range of scores and means across the different assessment methods. However, it is important to note that this offers insufficient evidence to suggest equivalence between the different competency assessment formats, and instead could indicate a plateau in achieving only moderate competence following completion of the training. Perhaps the measures are equivalent, yet the training program itself is insufficient in increasing participants' skills and knowledge, suggesting the need for additional supplemental training, supervision, and support ahead of care delivery.

It is noteworthy that in the pilot study [20] there was an observed improvement in competency on the multiple choice questionnaire from pre- to post-training, suggesting that while the training content appears to contribute to increased knowledge and skills related to HAP delivery, it may not be sufficient to ensure adequate knowledge of this treatment to deliver care. Thus, our results may show a ceiling effect with the training, since participants have not had the opportunity to practice their skills directly with patients in real world clinical settings. It will be necessary for NSP's who completed the training to begin delivering HAP in practice settings, as real-world experience enhanced with adequate support and ongoing supervision will likely further develop their skills and knowledge in treatment delivery. Re-assessing competency at later time points, using either mode (i.e., the validated multiple-choice questionnaire or standardized role-play assessment), can help to improve supervision efforts of the NSP's as they begin their delivery of depression care in routine care settings post-training. This would yield further opportunities to test both formats and collect data on specific skills the NSP's may need further training to support and assess whether total scores remain consistent or improve for the NSP's across both modes of assessment.

Increasing the capacity of the trained mental health workforce to meet the needs of the global burden of mental health problems requires scalable training programs complemented with scalable measures of competency assessment. The potential for scalability is an important consideration between the different approaches for competency measurement considered in this study. The strength of written assessments is that they are far more scalable because they are inexpensive and can be administered remotely using digital devices or via paper/pencil format. However, these written assessments provide limited insights about actual performance or ability to actively practice the skills in a live scenario with mock patients. Standardized and validated written measures could be easily administered as a first measure of competency assessment or as a standalone assessment in certain circumstances to assess the outcome of training to deliver HAP prior to any skill-based assessment using standardized role-plays assessments. While standardized role-plays offer an important approach to assessment of skill-based performance in a simulated professional practice setting, one of the biggest challenges surrounding this approach is the issue of scalability. Conducting role-plays requires considerable financial and logistical resources to implement and score, creating a bottleneck of outcomes for psychosocial training programs. Ongoing comparison of these two approaches, written and standardized role-plays, is needed to determine the consistency of outcomes across the two formats. Ultimately, though, determining provider skills for delivery of an evidence-based psychological intervention likely requires direct supervision and ongoing support in order to further develop skills, respond to challenging clinical encounters, and ensure high-quality care for patients [33, 34].

Limitations

Several limitations with the current study should be noted. First, a challenge with cultural adaptation was the requirement that technical terms used across the intervention are captured in the assessment tool, because oftentimes different terms or metaphors represent varying concepts and meanings across cultures. Second the 4-point Likert scale ratings on the standardized role-play measure were limited and did not provide much variation in scoring. This 4-point scale also did not provide any data on the NSP's' performance on specific competencies but instead the raters provided a global overall score for the assessment. Future work could use role-play rating scales that identify specific competencies so that the performance on these two modes of assessment could possibly be compared at the level of distinct competency as well as an overall global score. Third, the sample drawn for this pilot was small and therefore, was not adequately powered for any of the statistical analyses. Due to these limitations in rating the role-plays with a limited 4-point Likert scale as well as the small sample size, it was not possible to reach conclusions to support recommendations for selecting the optimal measurement tool, either between the multiple-choice assessment or the standardized role-play assessment. While our study showed consistent scores between the two modes of assessment, and a modest positive correlation between both of the role-plays and the multiple-choice assessment, additional work is needed to address these limitations. Future studies, using a larger sample, could perform more advanced rating, taking into consideration specific competencies that are consistent across both the multiple-choice assessment and the role-play assessments to allow for more advanced analysis to compare the modes of assessment in relation to specific competencies. Another key limitation is the

importance of linking competency assessment with quality-of-care delivery and patient outcomes, both key considerations necessary to address as part of future research.

Conclusion

When developed rigorously and with culture and context taken into consideration, competency assessment measures can serve multiple functions for quality and sustainable training and supervision programs. They can be an outcome of training for determining whether participants gain knowledge and skills following completion of the training content. Importantly, validated measures could potentially help identify gaps in providers' understanding of the content, and inform who may require additional training, support, or supervision. If results are consistent and reliable when various methods of competency assessment that are evaluating the same competencies are compared, implementers can then take several factors into consideration when deciding which method is more feasible, least resource intensive, assess the intended competencies, and is appropriate for use in the target setting and for supporting the target population.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s11126-022-10007-6.

Acknowledgements We acknowledge Aditya Anand and Minakshi Vishwakarma who provided support in data management and analysis. Lauren Mitchell who provided an overview of back translation, Abhijit Nadkarni and Steve Hollon who provided support at various stages in this research. Subhash Pednekar, Pranali Kundaikar, Lalan Karapurkar, and Suneina Haldankar who rated the role-play-based competency assessments Psychologists at Sangath Delhi who provided an overview of translations: Sanjana Guha and Sreepriya Menon.

Declarations

Ethics Approval All procedures performed in this study were approved by the ethics review boards at Sangath, India, and Harvard Medical School, United States, and are in accordance with the 1964 Helsinki declaration and its later amendments.

Informed Consent Informed consent was obtained from all individual participants included in the study.

References

1. Singla DR, Hollon SD, "The importance of implementation science in scaling up psychological treatments," *Behaviour Research and Therapy*, vol. 130. Elsevier Ltd, p. 103614, Jul. 01, 2020. https://doi.org/10.1016/j.brat.2020.103614
2. Barbui C, et al. Efficacy of psychosocial interventions for mental health outcomes in low-income and middle-income countries: an umbrella review. The Lancet Psychiatry. Feb. 2020;7(2):162–72. https://doi.org/10.1016/S2215-0366(19)30511-5
3. Singla DR, et al. Implementation and effectiveness of nonspecialist-delivered interventions for Perinatal Mental Health in High-Income Countries: a systematic review and Meta-analysis. JAMA Psychiatry. May 2021;78(5):498–509. https://doi.org/10.1001/JAMAPSYCHIATRY.2020.4556
4. Hoeft TJ, Fortney JC, Patel V, Unützer J, "Task-Sharing Approaches to Improve Mental Health Care in Rural and Other Low-Resource Settings: A Systematic Review," *J. Rural Heal*, vol. 34, no. 1, pp. 48–62, Dec. 2018. https://doi.org/10.1111/jrh.12229

5. Singla DR, Kohrt BA, Murray LK, Anand A, Chorpita BF, Patel V. Psychological treatments for the World: Lessons from low- and Middle-Income Countries. Annu Rev Clin Psychol. May 2017;13(1):149–81. https://doi.org/10.1146/annurev-clinpsy-032816-045217
6. Organization WH, "mhGAP Intervention Guide for Mental, Neurological and Substance Use Disorders in Non-Specialized Health Settings," World Health Organization, 2016. Accessed: Mar. 21, 2022. [Online]. https://www.ncbi.nlm.nih.gov/books/NBK390828/
7. Thornicroft G, et al. Undertreatment of people with major depressive disorder in 21 countries. Br J Psychiatry. Feb. 2017;210(2):119–24. https://doi.org/10.1192/BJP.BP.116.188078
8. Kohrt BA, Ramaiya MK, Rai S, Bhardwaj A, Jordans MJD, "Development of a scoring system for non-specialist ratings of clinical competence in global mental health: a qualitative process evaluation of the Enhancing Assessment of Common Therapeutic Factors (ENACT) scale," *Glob. Ment. Heal*, vol. 2, no. e23, pp. 1–16, 2015. https://doi.org/10.1017/GMH.2015.21
9. Kohrt BA, et al., "Ensuring Quality in Psychological Support (WHO EQUIP): developing a competent global workforce," *World Psychiatry*, vol. 19, no. 1, pp. 115–116, Feb. 2020. https://doi.org/10.1002/wps.20704
10. Cooper Z, et al. Assessing therapist competence: development of a performance-based measure and its comparison with a web-based measure. JMIR Ment Heal. Oct. 2017;4(4):e. 51. https://doi.org/10.2196/mental.7704
11. Kaslow NJ, Grus CL, Campbell LF, Fouad NA, Hatcher RL, Rodolfa ER. Competency Assessment Toolkit for professional psychology. " Train Educ Prof Psychol. 2009;3:no. 4S, p. S27.
12. Pilnick A, Trusson D, Beeke S, O'Brien R, Goldberg S, Harwood RH. "Using conversation analysis to inform role play and simulated interaction in communications skills training for healthcare professionals: identifying avenues for further development through a scoping review 08 information and Computing Sciences 0801 Artificial Intelligence and Image Processing," BMC Med Educ, 18, 1, Nov. 2018. https://doi.org/10.1186/S12909-018-1381-1
13. Kohrt BA, et al., "Therapist competence in global mental health: Development of the ENhancing Assessment of Common Therapeutic factors (ENACT) rating scale," *Behav. Res. Ther*, vol. 69, pp. 11–21, Jun. 2015. https://doi.org/10.1016/j.brat.2015.03.009
14. Restivo JL, et al. Assessing health worker competence to deliver a brief psychological treatment for depression: development and validation of a scalable measure. J Behav Cogn Ther. Oct. 2020. https://doi.org/10.1016/j.jbct.2020.10.001
15. Naslund JA, et al. Digital training for non-specialist health workers to deliver a brief psychological treatment for depression in India: protocol for a three-arm randomized controlled trial. Contemp Clin Trials. Jan. 2021;102:106267. https://doi.org/10.1016/j.cct.2021.106267
16. Patel V, et al. The healthy activity program (HAP), a lay counsellor-delivered brief psychological treatment for severe depression, in primary care in India: a randomised controlled trial. Lancet. 2017;389(10065):176–85.
17. Fairburn CG, Cooper Z. Therapist competence, therapy quality, and therapist training. Behav Res Ther. 2011;49:no. 6–7, pp. 373–378. https://doi.org/10.1016/j.brat.2011.03.005
18. Sangraula M, et al., "Development of the mental health cultural adaptation and contextualization for implementation (mhCACI) procedure: a systematic framework to prepare evidence-based psychological interventions for scaling," Glob Ment Heal, 8, 2021. https://doi.org/10.1017/GMH.2021.5
19. Asher L, et al., "'Like a doctor, like a brother': Achieving competence amongst lay health workers delivering community-based rehabilitation for people with schizophrenia in Ethiopia," *PLoS One*, vol. 16, no. 2, p. e0246158, Feb. 2021. https://doi.org/10.1371/JOURNAL.PONE.0246158
20. Muke SS, et al., "Digital training for non-specialist health workers to deliver a brief psychological treatment for depression in primary care in India: Findings from a randomized pilot study," *Int. J. Environ. Res. Public Health*, vol. 17, no. 17, pp. 1–22, Sep. 2020. https://doi.org/10.3390/ijerph17176368
21. Ministry of Health & Family Welfare Government of India. "New Pathways, New Hope - National Mental Health Programme (NMHP)," 2014. https://nhm.gov.in/index1php?lang=1&level=2&sublinkid=1043&lid=359 (accessed Mar. 31, 2022).
22. Khan A, et al., "Design and Development of a Digital Program for Training Non-Specialist Health Workers to Deliver an Evidence-Based Psychological Treatment for Depression in Primary Care in India," *J. Technol. Behav. Sci*, vol. 5, no. 4, pp. 402–415, Dec. 2020. https://doi.org/10.1007/S41347-020-00154-7
23. Harkness JA, "Round 4 ESS translation strategies and procedures," *Round 4 ESS Transl. Strateg. Proced*, pp. 1–20, 2008.
24. Behr, D., & Shishido, K. (2016). The translation of measurement instruments for cross-cultural surveys. In C. Wolf, D. Joye, T. W. Smith, & Y. Fu (Eds.), *The SAGE handbook of survey methodology* (pp. 269?287). London: Sage.
25. Harkness JA, Villar A, Edwards B. "Translation, adaptation, and design," Surv Methods Multicult Multinatl Multiregional Context, 115–40, May 2010. https://doi.org/10.1002/9780470609927.CH7
26. Harkness JA, Dinkelmann K, Pennell B-E, Mohler PP. "Final report and translation tool specifications (Deliverable No. 5)," Eur. Soc. Surv. Infrastruct. Proj. Improv. Soc. Meas. Eur., 2007.

27. Patel V, et al., "The effectiveness and cost-effectiveness of lay counsellor-delivered psychological treatments for harmful and dependent drinking and moderate to severe depression in primary care in India: PREMIUM study protocol for randomized controlled trials.," *Trials*, vol. 15, no. 1, p. 101, Apr. 2014. https://doi.org/10.1186/1745-6215-15-101
28. Singla DR, et al. Improving the scalability of psychological treatments in developing countries: an evaluation of peer-led therapy quality assessment in Goa, India. Behav Res Ther. 2014;60(100):53–9. https://doi.org/10.1016/j.brat.2014.06.006
29. Ottman KE, Kohrt BA, Pedersen GA, Schafer A. Use of role plays to assess therapist competency and its association with client outcomes in psychological interventions: a scoping review and competency research agenda. Behav Res Ther. 2020. https://doi.org/10.1016/j.brat.2019.103531
30. Resnicow K, Baranowski T, Ahluwalia JS, Braithwaite RL, "Cultural sensitivity in public health: Defined and demystified," *Ethn. Dis*, vol. 9, no. 1, pp. 10–21, Dec. 1999.
31. Chu J, Leino A. Advancement in the maturing science of cultural adaptations of evidence-based interventions. J Consult Clin Psychol. Jan. 2017;85(1):45–57. https://doi.org/10.1037/CCP0000145
32. Heim E, Kohrt BA. "Cultural adaptation of scalable psychological interventions: a new conceptual framework," Clin Psychol Eur, 1, 4, Dec. 2019. https://doi.org/10.32872/CPE.V1I4.37679
33. Singla DR, Ratjen C, Krishna RN, Fuhr DC, Patel V. "Peer supervision for assuring the quality of non-specialist provider delivered psychological intervention: Lessons from a trial for perinatal depression in Goa, India," 2019. https://doi.org/10.1016/j.brat.2019.103533
34. Singla DR, Kumbakumba E. The development and implementation of a theory-informed, integrated mother-child intervention in rural Uganda. Soc Sci Med. Dec. 2015;147:242–51. https://doi.org/10.1016/J.SOCSCIMED.2015.10.069
35. Sha, M., & Pan, Y. (2013). Adapting and improving methods to manage cognitive pretesting of multilingual survey instruments. *Survey Practice, 6*(4). Retrieved from http://www.surveypractice.org/index.php/Surv

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